EEE

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), Jul-Dec 2021

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

1. (a). Explain how transistor is used as inverter? Draw the circuit of a CMOS switch. 3+2
   (b). Discuss on speed of operation, noise immunity and figure of merit in digital logic families. 3
   (c). Draw the circuit diagram of 3 inputs RTL NOR gate and explain how it works. 6

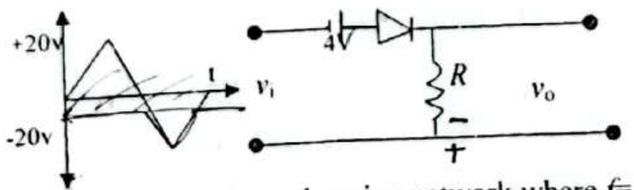
2. (a). For the DTL NAND gate where diode and transistors parameters are given as
   Diode: Voltage across the diode=0.7V; Cut-in voltage=0.6V
   Transistor: Cut-in voltage=0.5V; $V_{BE.sat}$=0.8V ; $V_{CE.sat}$=0.2V ; h_{FE}=20; R_C= 2 KΩ; R_B= 4 KΩ
   Resistor connected with V_{cc}=4 KΩ. Calculate i) fan-out ii)noise-margins iii) average power- *P*. 6
   (b). How 3-input TTL NAND gate is operated in digital logic families? 4
   (c) Discuss about the saturated bipolar logic families. 4

3. (a) Show the pin configuration of a typical op-amp with its function. 4
   (b) Draw the circuit of an Op-Amp as voltage follower and find an expression for its voltage gain. 6
   (c) Draw the circuit for a non-inverting amplifier and explain its operations. 4

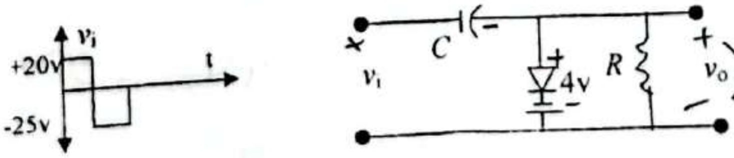
4. (a) Show the differences between open loop gain and close loop gain. 3
   (b) Draw the circuit diagram of integrator and differentiator and explain how it works. 6
   (c) Draw the circuit diagram of subtractor and explain how it works. 5

5. a) Find the analog output voltage of a 3-bit D/A converter for all possible inputs. Assume *K*=1 5
   b) Design a weighted-resistor D/A converter and explain how it works. 5
   c) How offset voltage is used in D/A converter? 2
   d) What is quantization error? 2

6. a) Analyze the clipper network with given input and determine the output waveform of the given network. 6



b) Analyze output voltage for given clamping network where *f*= 2500 Hz, *R*= 25 Kilo-Ohm, *C*= 10 nF?



c) How resistors and capacitors are considered for clampers?

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Answer any 5 of the following questions

1. (a). Discuss the switching characteristics of a semiconductor diode. 4
   (b). Draw the circuit diagram of 3-input RTL NOR Gate. Write down the working principle of the 3-input RTL NOR Gate. 6
   (c). Calculate the noise margin of the RTL NOR gate if the parameters are considered as V_{cc}= 5.0; R_L=360Ω ; R_B=450Ω ; Cut in voltage of transistor =0.5 V; N=5; h_{FE}= 10. 4

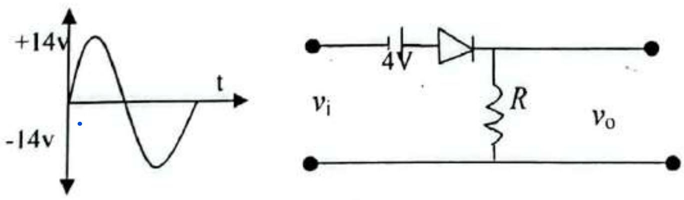
2. (a). Draw the circuit of a 3-input TTL NAND gate and explain how it works. 6
   (b). Show the working principle and applications of CMOS inverter. 4
   (c) Discuss the speed of operation and fan out of logic gates. 4

3. (a) Show the circuit symbol and terminals of operational amplifier 741C. 4
   (b) Show the block diagram for using op-amp in sensor based data acquisition. 4
   (c) Draw the circuit for an inverting amplifier. Deduce the equation of output voltage, current and close lop gain for inverting amplifier. 6

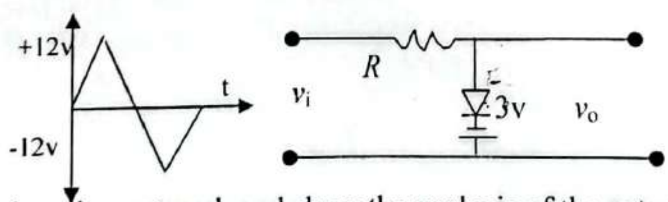
4. (a). Explain how an operational amplifier can be used to design voltage follower? 4
   (b). Show the circuit diagram of integrator and differentiator and explain how it works. 6
   (c) How an operational amplifier can be used to build inverting adder? 4

5. a) Design a 4 bit unipolar D/A converter with V(1)= -1, V(0)=0V and R_F =8R. Find the analog output voltage of the D/A converter for each of the digital inputs from 0000 to 1111. 5
   b) Design a 3 bit parallel-comparator A/D converter and explain how it works. 5
   c) What is quantization error? How it is reduced in A/D converter? 4

6. a) Analyze the clipper network with given input and determine the output waveform of the given network. 6



   b) Determine the output waveform step by step of the given network. 4



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   c) Design the clamping network and show the analysis of the network. 4

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), July-Dec 2019

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

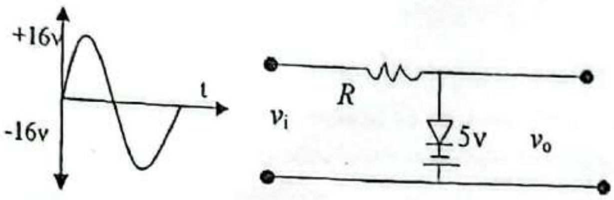
1. (a). Discuss the bipolar logic families for electronics devices. 4

(b). Draw the circuit diagram of 3-input DTL NAND Gate. Calculate fan out, noise margin and average power if the parameters are considered as voltage across a conducting diode = 0.7 V; Cut in voltage of diode = 0.6 V; Cut in voltage of transistor = 0.5 V; $V_{BE,sat}$= 0.8; $V_{CE,sat}$= 0.2; h_{FE}= 20. 7

(c). Show the limitations of DTL and TTL circuits. 3

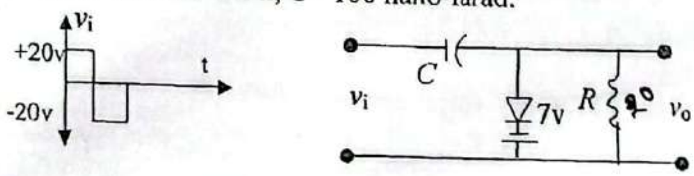
2. (a). Draw the circuit of a TTL gate with totem-pole output driver. Write down the working principle of the circuit. 6

(b). Determine the output waveform step by step of the given network. 6



(c) How figure of merit is measured for a digital IC? 2

3. (a) Analyze output voltage and time constant is analyzed in given clamping network where f= 2000 Hz, R= 20 Kilo-Ohm, C= 100 nano-farad. 5



(b). What are differences between open loop gain and close loop gain of operational amplifier? 3

(c). Show how can an operational amplifier be used as an adder and subtractor? 6

4. (a). Explain how an operational amplifier can be used as an integrator? 5

(b). Show how a low pass filter is designed using the operational amplifier? 5

(c) How voltage gain is obtained for inverting amplifier? 4

5. (a). Draw the circuit diagram of a multichannel amplifier. Show the output voltage analysis and circuit diagram of differentiator. 6

(b). Explain how an op-amp can be used as voltage follower? 4

(c) Give the examples where D/A and A/D converters are used. 4

6. (a). Deduce the output voltage for the weighted-resistor D/A converter. Why offset voltage is used for D/A converter? 3+2

(b). Find the analog output voltage of a 5-bit D/A converter where K=1. Design the D/A converter for this. 10·1 4 6

(c) How 4 bit digital output can be obtained from an analog voltage in the range of 0 to V? 3

Patuakhali Science and Technology University

7^{th} Semester (L-4, S-1), Final Exam. of B. Sc. Engg. (CSE), July-Dec 2018

Course Code: CSE-413 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

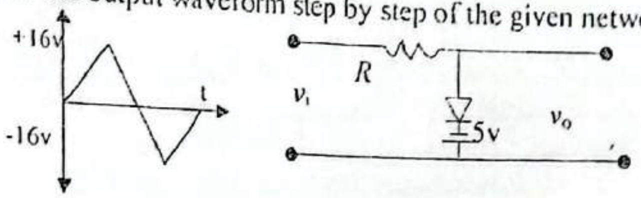
1. (a). Write down the short note on i) Figure of Merit ii) Fan out iii) Noise immunity. 6

(b). Draw the circuit diagram of 3-input DTL NAND Gate. Explain the operation of DTL NAND gate. 5

(c). Show the limitations of RTL and DTL circuits. 3

2. (a). Why clipper is used in electronic circuits? 2

(b). Determine the output waveform step by step of the given network



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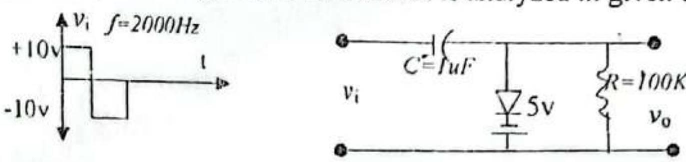
(c). Analyze output voltage and time constant is analyzed in given clamping network?



6

3. (a). Write down the characteristics of ideal op-amp. 3

(b). What is open-loop voltage gain and closed-loop gain? Show the pin configuration of a typical op-amp. 5

(c). How output voltage is calculated for the circuit of (i) non-inverting amplifier ii) summing amplifier? 6

4. (a). How cascaded op-amp circuit is designed? How output voltage is obtained in cascaded op-amp circuit? 7

(b). Design an op-amp circuit with inputs v_1 and v_2 such that $v_o = -7v_1 + 3v_2$ 4

(c). How voltage gain is obtained for inverting amplifier? 3

5. (a). Show the output voltage analysis and circuit diagram of differentiator. 5

(b). Explain why capacitor and inductor are very useful in electric circuits. 4

(c) Design an analog computer circuit to solve the differential equation

$$\frac{d^2v_o}{dt^2} + 3\frac{dv_o}{dt} + 2v_o = 4\cos 10t, \quad t>0 \text{ subject to } v_o(0)=2, \ v_o'(0) = 0$$ 5

6. (a). Analyze the output voltage and K factor of the weighted-resistor D/A converter. 4

(b) Consider a 4bit D/A converter with $V(1)$= -1V, $V(0)$=0V and R_F=16R. Obtain the analog output voltage for each digital input where V_o=0V for the digital input of 1000. Draw the circuit of D/A converter and adjust the offset voltage for the circuit. 6

(c) What is quantization error? How quantization error can be reduced in analog to digital converter? 4

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Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), July -Dec. 2016

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Draw a 3-input DTL NAND gate. Show the performance of DTL NAND gate based on your drawn circuit. [Considering standard values of diode and transistor parameters; h_{FE}=40] 6

(b). Discuss about the staurated bipolar logic families. 4

(c). Find out the logic 0 and logic 1 noise margins for the given circuit diagram of Fig. 1(c) where h_{FE}=10 and N=5. 4

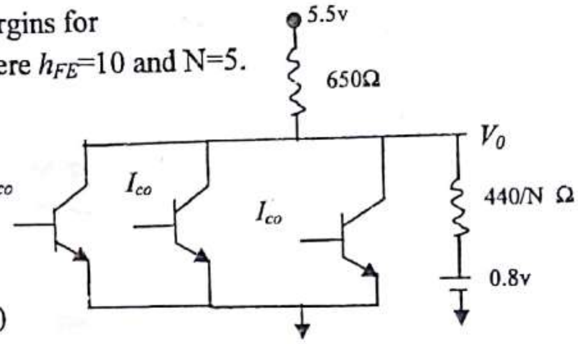


Fig.1(c)

2. (a). Why TTL is the most popular general purpose logic family? 3

(b). If the digital input to the D/A converter is $\overline{b_3}b_2b_1b_0$, find the analog output voltage such that V=0 for a digital input of 0000. Design a D/A converter for the mentioned criteria. 3+4

(c). How quantization error can be reduced in A/D converter? 4

3. (a). Determine V_0 for the given network of Fig. 3(a). 6

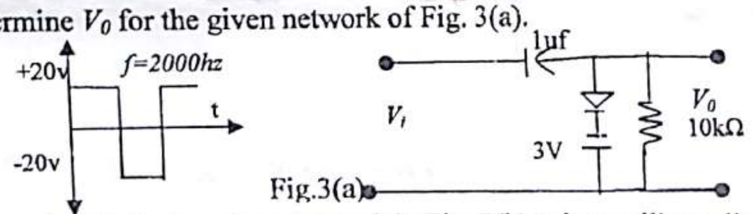


Fig.3(a)

(b). Determine V_0 for the given network in Fig. 3(b) using a silicon diode. 5

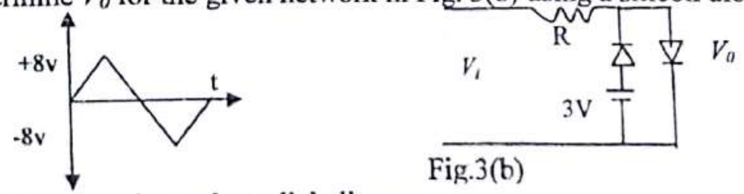
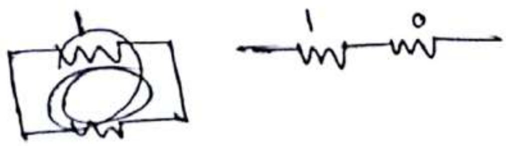


Fig.3(b)

(c). Distinguish series and parallel clippers. 3

4. (a). What is an operational amplifier? "The inverting amplifier provides a constant voltage gain with 180° phase shift"- justify the statement with relevant circuit diagram. 5

(b). Define slew rate and active device. Solve the differential equation using appropriate circuit diagram: $D^3y - 4D^2y + 6Dy - 8y = -f(t)$ volts where $D = d/dt$ 5



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(c) Calculate the output voltage (V_{out}) for the circuit of fig.4(c)

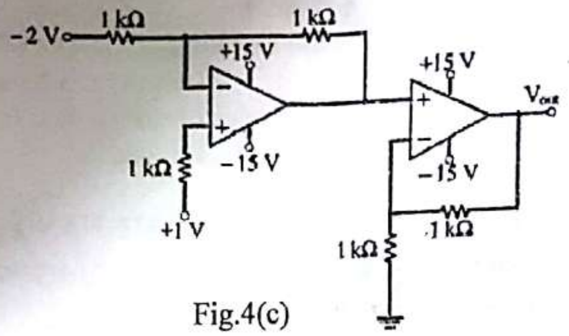

Fig.4(c)

5. (a). Write down the advantage of active filter. Design a 70 kHz with 50% duty cycle square wave generator using 555. 5

(b) Sketch the equivalent circuit of an ideal Op-Amps. Suppose for a non-inverting amplifier, the equivalent circuit of an Op-Amps is shown in figure 5(b). Find: i) $Vs/Vout$ for R_{in}=100kΩ. ii)$Vs/Vout$ for an ideal op-amp. 9

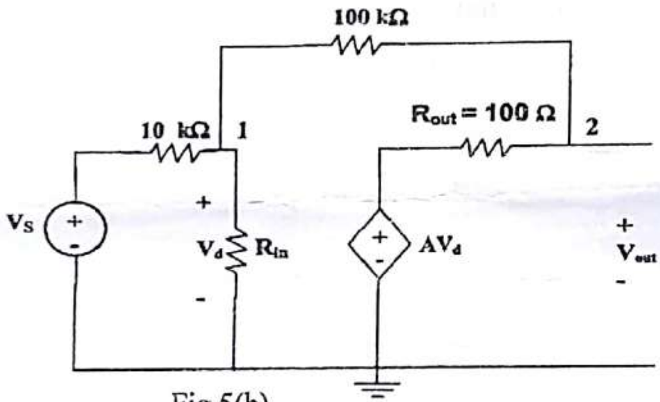

Fig.5(b)

6. (a). What do you mean by cut off frequency? Show that in case of 1st order high pass filter, the transfer function is equal to the 70.7% of it's voltage gain. 5

(b). "Monostable multivibrator is known as one-shot circuit" – Justify the statement with relevant circuit diagram. 4

(c). Explain the functional block diagram of 555 timer. 5

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A	B	C	OU
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Patuakhali Science and Technology University
6^{th} Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), July -Dec. 2015
Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique
Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours
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Answer any 5 of the following questions:

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1. (a). Explain the operation of a 3-input RTL OR gate with its circuit diagram. 6
(b). Write down the short note on i. Figure of merit ii. Noise Margins. 4
(c). Find out the logic 0 and logic 1 noise margins for the given circuit diagram of Fig. 1(c). 4

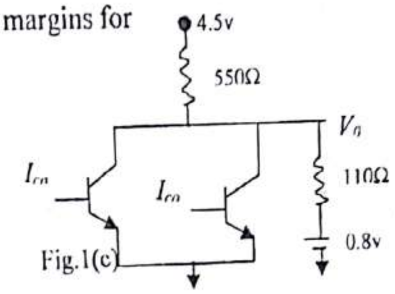

Fig.1(c)

2^3-1

2. (a). Derive the equation of V_0 for weighted resistor D/A converter. 5
(b). What are the problems of weighted resistor D/A converter? 3
(c). Consider a 3-bit digital input in 1's complement format. Design a D/A converter for this. 6

3. (a). How clamping network is analyzed? Determine V_0 for the given network of Fig. 3(a). 9

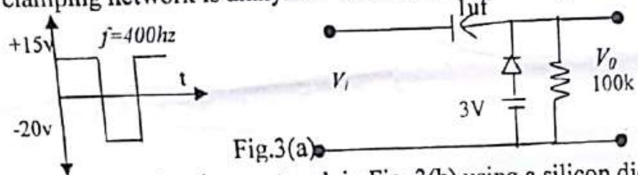

Fig.3(a)

$V_o = V_{in} + 3V$
$= 20 + 3V$. 23V

(b). Determine V_0 for the given network in Fig. 3(b) using a silicon diode with $V_T=0.7V$; 5

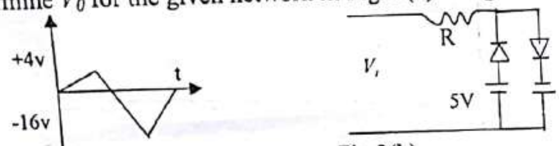

Fig.3(b)

4. (a). What is an operational amplifier? What are the fundamental differences between ideal & practical Op-Amp (Model 741c)? 4
(b). Solve the differential equation using appropriate circuit diagram: 4
$$D^3y + 4D^2y - 6Dy - 8y = -5 \text{ volts where } D = \frac{d}{dt}$$
(c). Write down the short notes on: i. Op-Amp is a subtractor ii. Frequency scaling iii. Schmitt trigger. 6

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Encoder

5. (a). Design a 50 kHz, 70% duty cycle square wave generator using 555 timer. 5
(b). Show that in case of 1^{st} order low pass filter, the transfer function is equal to the 70.7% of it's voltage gain. 5
(c). For the differentiator circuit in the Fig.5(c), determine the output voltage and sketch the output waveform. 4

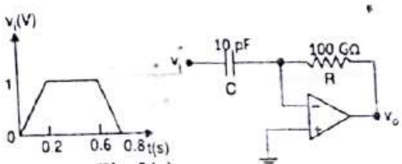

Fig.5(c)

$V_o = \int \frac{1}{RC} V_i dt$
$V_o = RC \frac{dv}{dt}$

6. (a). Evaluate the time period for a free running multivibrator & show that the frequency is independent of output voltage of the square wave. 5
(b). "Monostable multivibrator is known as one-shot circuit" – Justify the statement with relevant circuit diagram. 4
(c). Explain the functional block diagram of 555 timer. 5

RC
$\int \frac{1}{RC} V_{in}$

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E3

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), July -Dec. 2014

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

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Answer any 5 of the following questions:

1. (a). Explain the operation of a 3-input TTL NAND gate with its circuit diagram. 6
   (b). Explain the operation of 2-input CMOS OR gate 4
   (c). In the RTL 2-unput NOR gate, $V_{cc}=5V$, all resistances are 640Ω. Calculate the average power supply when it is driving 4 gates. Assume $V_{BE,Sat}=0.8V, V_{CS,Sat}=0.2V, h_{FE}=20$. 4

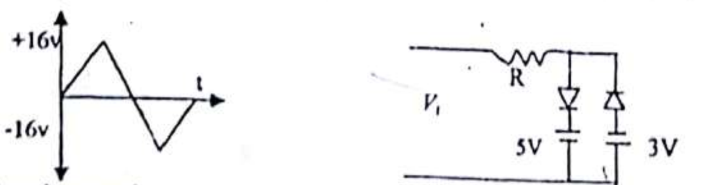
2. (a). Discuss the operations of a monostable multivibrator with its circuit diagram where a trigger input is given at a NAND gate. 7
   (b). Draw a circuit diagram for a comparator where reference voltage $V_{ref}=5V$ and input voltage $V_{in}=10\sin 4\pi t$. Show the input and output waveforms for the given circuit. 7

3. (a). Draw the functional block diagram of 555 timer. Explain its operations. 8
   (b). Discuss the function of a monostable multivibrator using 555 timer 6

4. (a). Design a D/A converter with $V(1)=-2V$, $V(0)=0V$ and $R_F=4R$ for 5 bit digital input from 00000 to 11111 7
   (b). Adjust the offset voltage for the D/A converter such that $V_0=0$ for digital input of 01000. 7

5. (a). Design a 3-bit parallel comparator A/D converter. Explain the operation of this A/D converter. 8
   (b). Explain how the quantization error can be reduced for this A/D converter. 6

6. (a). Determine V_0 for the given network using a silicon diode with $V_T=0.7V$; 6



   (b). Why clamper is used in the electronic circuit? Determine V_0 for the given network 8